

Length of an arc on the surface

Def. Let  $X$  be a patch and

$$\alpha: [a, b] \rightarrow \mathbb{R}^3: t \mapsto X(u(t), v(t)),$$

then the arc length is

$$L(\alpha) = \int_a^b |\alpha'(t)| dt = \int_a^b \sqrt{I_{\alpha(t)}(\alpha'(t))} dt$$

Area of Torus

$$\text{Area}(X(B)) = \iint_B \sqrt{EG - F^2} \, du \, dv.$$

Consider the torus: for  $0 < \phi, \theta < 2\pi$ ,

$$X(\phi, \theta) = ((a + r \cdot \cos \theta) \cdot \cos \phi, (a + r \cdot \cos \theta) \cdot \sin \phi, r \cdot \sin \theta)$$

so that

$$X_\phi = (-(a + r \cos \theta) \sin \phi, (a + r \cos \theta) \cos \phi, 0)$$

$$X_\theta = (-r \sin \theta \cos \phi, -r \sin \theta \sin \phi, r \cos \theta)$$

$$\text{Now, } E = (a + r \cos \theta)^2,$$

$$F = 0$$

$$G = r^2.$$

Now,

$$\text{Area} = \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^{2\pi - \epsilon} \int_{\epsilon}^{2\pi - \epsilon} (a + r \cos \theta) \, d\phi \, d\theta.$$